Understanding How Cypress Works

Although the Cypress test seems simple, Cypress is a tool where the apparent simplicity of the developer commands hides a fair amount of complexity behind the scenes. In order to effectively write tests in Cypress, it is important to take a step back to understand how Cypress works to avoid a lot of confusion later on. Cypress's asynchronous structure means that a lot of common JavaScript patterns either won't work or aren't recommended in Cypress. For example, using regular variable assignment is going to cause you trouble.

The most important thing to understand about Cypress is that although Cypress commands appear to be regular JavaScript function calls, in fact all Cypress commands are asynchronous. In effect, each time you use the `cy` command, you are creating something like a JavaScript promise, and subsequent `cy` commands are in the `then` clause of the preceding command, meaning that each successive `cy` command only executes after the previous one has completed.

When the Cypress command is actually run as part of the test, it queues a list of commands in the test and returns immediately. What we think of as the actual command—the `get` or `click` or whatever—only executes as part of this queue of promises once the entire test has loaded.

The most important implication of this behavior is that Cypress is a world unto itself. Normal JavaScript assignment or logic that is not mediated through Cypress does not see Cypress stuff at all, and conversely, everything you do in a Cypress test needs to go through the `cy` command. Cypress has constructs to allow you to do things like variable assignment and logic inside Cypress-land, but if you write normal variable assignments with `let` or `const`, those assignments will happen as the test loads. They will not be able to see Cypress data, and Cypress commands won't be able to see those variables.

I do need to point out here that Cypress commands are not exactly the same as JavaScript promises and you can't directly mix the two. Specifically, using `async/await` in a Cypress test will not put your async code in the Cypress chain of